Problem Statement: Inefficiencies and Risks in Manual Examination Attendance Verification

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1. Background

At the University of Eswatini, examination attendance is currently verified through the use of **physical examination cards** issued to students for each course they are registered to write. During the examination session, these cards are signed by students and then collected by invigilators as proof of attendance.

This manual system plays a critical role in confirming that a student has sat for a particular exam. However, the process is heavily dependent on physical documentation, which introduces several operational and security challenges.

2. Problem Description

Main Problems:

Risk of Loss of Examination Cards: Examination cards can easily be lost, misplaced, or damaged after collection. Once lost, there is no reliable backup to confirm whether a student attended the exam.

Manual and Error-Prone Counting Process: Invigilators are required to manually count the collected examination cards to determine attendance. This process is time-consuming and prone to human error, especially in large exam venues.

3. Impact of the Problems

These challenges lead to:

- Reduced confidence in the integrity of the examination process
- Increased administrative workload for invigilators and academic staff
- Potential disputes over student attendance and script ownership
- Delays in resolving examination-related issues

4. Need for a Solution

There is a clear need for a more secure, efficient, and reliable system for verifying examination attendance. An improved solution should:

- Provide a tamper-proof record of student attendance
- Enable accurate and automated attendance tracking
- Reduce reliance on physical documents
- Improve accountability, traceability, and data availability
- Minimize manual effort and human error

5. Objective of the Project

The objective of this mini project is to explore and design a digital or technology-driven solution that addresses the limitations of the current manual examination card system, while improving efficiency, security, and reliability in authenticating student before entering the exam room and during examinations.